# pulp protection

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***Pulp protection*** in dentistry, means the strategies used to protect the dental pulp (the soft tissue inside the tooth containing nerves and blood vessels) from damage during restorative procedures like cavity preparation or trauma. The goal is to prevent pulp inflammation, infection, or necrosis.

## Types of Pulp Protection

Pulp protection can be divided into **direct** and **indirect** pulp capping.

*Pulp capping* is defined as "endodontic treatment designed to maintain the vitality of the endodontium." Several favorable conditions must be present before considering direct or indirect pulp capping:

- The tooth must have a vital pulp and no history of spontaneous pain.
- Pain elicited during pulp testing with a hot or cold stimulus should not linger after stimulus removal.
- A periapical radiograph should show no evidence of a periradicular lesion of endodontic origin.
- Bacteria must be excluded from the site by the restoration

## A. Direct Pulp Capping

- *Direct pulp capping* is an attempt to maintain pulpal vitality by placing a material directly over the exposed pulp. It is hoped that this will allow the pulp to heal normally, regenerate reparative dentin, and prevent the need for more extensive and expensive treatment, such as root canal therapy.

- Direct pulp capping is more likely to be successful if the cause of the pulp exposure is mechanical rather than due to caries. A carious exposure will cause bacterial contamination of the pulp resulting in inflammation and a pulp that is less able to respond and heal, whereas mechanical exposure will likely cause less bacterial contamination and resulting inflammation.

- After a pulp has been exposed, it is important to control pulpal bleeding before placing a pulp capping agent. It is normally controlled with a cotton pellet saturated with water or saline applied to the exposure site.

## B. Indirect Pulp Capping

In which carious dentin (affected dentin) is left over a vital pulp to avoid pulp exposure and is covered with a biocompatible material then permanent restoration is applied in the same visit to improve success.

- Clinical evaluation at 6–12 months by pulp testing and radiographic monitoring.
- Used when the carious lesion is very deep but the pulp is not yet exposed to protect the pulp and promote secondary dentin formation.

## Materials for Pulp Protection

### 1. Calcium hydroxide

It has been considered the gold standard of direct pulp capping materials for several decades because of its excellent antibacterial properties and high pH that causes irritation of the pulpal tissue, which stimulates repair via some unknown mechanism leading to the formation of dentin bridge.

Calcium hydroxide has some disadvantages attributed to its use as pulp capping material include:

- The self-curing formulations are highly soluble and are subject to dissolution over time.
- It has no inherent adhesive qualities and provides a poor seal.

### 2. Mineral trioxide aggregate (MTA)

Is a bioactive (release Calcium ions) endodontic cement widely used for pulp therapy and root repair because of its excellent biocompatibility, sealing ability, and ability to stimulate hard tissue formation.


## Composition

- Tricalcium silicate
- Dicalcium silicate
- Tricalcium aluminate

- Calcium sulfate (small amounts)
- Bismuth oxide (radiopacifier)

* When mixed with water, it forms a calcium silicate hydrate gel and releases calcium hydroxide, which contributes to its bioactivity.

* It is available in two colors—white and gray color. The gray version is created by the addition of iron.

## Properties

- Excellent sealing ability
- Highly biocompatible
- Induces dentin bridge formation
- Antibacterial (due to high pH ≈ 12.5)
- It sets in a moist environment (hydrophilic in nature).

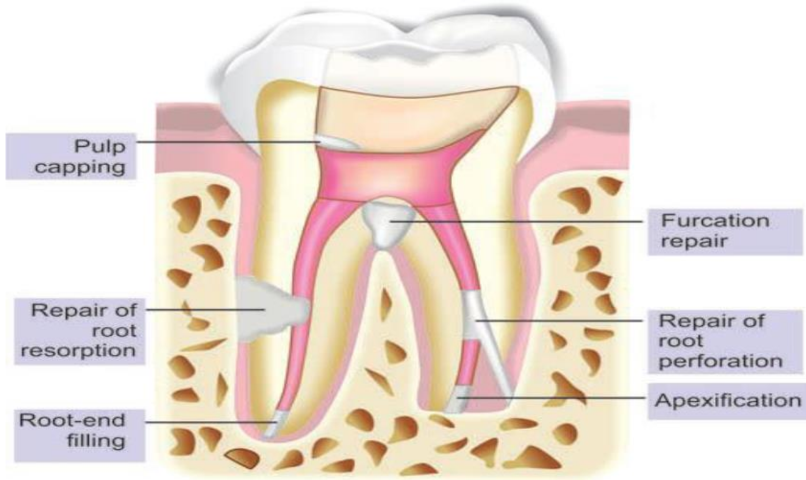
## Drawbacks

- Long setting time (≈ 2 hours and 45 minutes)
- Difficult handling
- The presence of iron in the gray MTA formulation may darken the tooth

## Manipulation of MTA

To prepare MTA, a small amount of liquid (sterile water) and powder are mixed to putty consistency (moist sand appearance). Since, MTA mixture is a loose granular aggregate, it does not stick very well to any instrument. It cannot be carried out in cavity with normal cement carrier and thus has to be tried with amalgam carrier or especially designed carrier (MAP system). Once MTA is placed, it is compacted lightly with moist burnishers and micro pluggers.

**Indications of MTA**




## 3. Biodentine

It is a tricalcium silicate–based material supplied as capsule (powder) + liquid ampoule (calcium chloride accelerator + water-reducing agent) mixed by adding the entire liquid ampoule to the powder capsule then triturate in amalgamator for 30 seconds. It has faster setting time (~12 minutes) with better biologic, handling and physical properties, superior sealing property than MTA.


## 4. Resin-Modified Calcium Silicate Materials e.g. TheraCal LC

It is a light-cured Calcium-releasing material therefore it may cause heat generation with some concerns regarding resin components and pulp response.

* It comes in a one paste system

* Have an instant high Ca release for short term, while dycal has an instant low Ca release for long term

* Lower calcium ion release compared to pure calcium silicates.


TheraCal™ LC